<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>5</td><td>(a)</td><td></td><td></td><td>As the water is heated {it expands / its volume increases} and density decreases (1)
The cooling water’s volume decreases and its density increases (1)

Or
density decreases when [water] heated and density increases when it cools(1)..
.. because {it / water} expands when heated and contracts when cools (1)

Not: <u>particles</u> expand or become less dense</td><td>2</td><td></td><td></td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>Indicative content:
Ion/lattice process
1. the molecules/ions at the “high temperature end” of the substance gain energy
2. and vibrate with increasing amplitude / kinetic energy.
3. They transfer their energy to other neighbouring molecules/ions by collisions and this continues through the body of the material.
Free-electron process
4. Copper has free (or de-localised or mobile) electrons which gain energy at the “high temperature end”
5. and move / migrate towards the “lower temperature end”,
6. transferring energy more quickly

5 – 6 marks
Expect at least 4 statements with at least two from each process
A correct, description of the two processes that identify metals as good conductors of thermal energy is given.
There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</td><td>6</td><td></td><td></td><td>6</td><td></td><td>6</td></tr>
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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				3 – 4 marks Expect at least 1 statement from each process A reasonable attempt is made at describing both processes OR a totally correct, detailed description of one of the processes is given. The candidate typically uses terms like ”vibrates more”. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Expect any statement from either of the processes An attempt is made at explaining conduction in terms of particle movement without the identity of the particles being stated – usually of the vibration of molecules in which the candidate typically uses the term “the molecules start to vibrate…” *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*. *The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 5 total	**8**	**0**	**0**	**8**	**0**	**8**